

Epidemiological Surveillance Methodology

1. GENERAL

SCHOOL	PUBLIC HEALTH		
SECTION	PUBLIC HEALTH POLICIES		
LEVEL OF STUDIES	MASTER'S DEGREE IN PUBLIC HEALTH		
COURSE CODE	MSPYL302	SEMESTER OF STUDY	C
COURSE TITLE	EPIDEMIOLOGICAL SURVEILLANCE METHODOLOGY		
INDEPENDENT TEACHING ACTIVITIES <i>in case the credits are awarded in discrete parts of the course e.g. Lectures, Laboratory Exercises, etc. If credits are awarded uniformly for the whole course enter the weekly teaching hours and the total credits</i>		WEEKLY HOURS OF TEACHINGSKALIAS	CREDIT UNITS
Lectures		2	
Laboratory Exercise		1	
Add rows if needed. The organisation of teaching and the teaching methods used are described in detail in 4.		3	6
COURSE TYPE <i>Background, General Knowledge, Scientific Area, Skills Development</i>	Specialization course Scientific area, Skills Development		
PREREQUISITE COURSES:	None		
LANGUAGE OF TEACHING AND EXAMS:	Greek		
THE COURSE IS OFFERED TO ERASMUS STUDENTS	Offered in Greek		
ELECTRONIC COURSE PAGE (URL)	www.moodle.uniwa.gr		

2. LEARNING OUTCOMES

Learning Outcomes <i>The learning outcomes of the course are described, the specific knowledge, skills and abilities of an appropriate level that the students will acquire after the successful completion of the course.</i> <i>Consult Annex A</i> • Description of the Level of Learning Outcomes for each cycle of study according to the Qualifications Framework of the European Higher Education Area • Descriptive Indicators of Levels 6, 7 & 8 of the European Qualifications Framework for Lifelong Learning and Annex B • Concise Guide to Writing Learning Outcomes
After the successful completion of the course the student will be able to: • understand the content and characteristics of the epidemiological crisis. • allocate the main objectives of epidemiological surveillance systems and distinguish their main species. • identify the basic principles and characteristics of the operation of the epidemiological surveillance systems in Greece. • describe the characteristics of organisation, design and evaluation of epidemiological surveillance systems. • be aware of the objectives and specific characteristics of epidemiological surveillance in emergency situations. • understand the role of the laboratory and the possibilities offered by the use of new technologies in epidemiological surveillance. • understand modern approaches to epidemiological surveillance, its dynamic evolution as well as the possibilities offered by the use of new technologies. • participate in the epidemiological surveillance system design team for a given disease under defined conditions.

- understand the objectives and phases of investigation of an epidemic and understands the organizational issues and the framework of the required collaborations when investigating an epidemic.
- know the different types of epidemics and their characteristics and can describe an epidemic in terms of place, time and characteristics of individuals.
- distinguish the basic principles in the design and conduct of cohort studies and case-control studies in the context of investigating an epidemic.
- calculate and interpret the main frequency and correlation measures needed to investigate an epidemic.
- actively participate and work effectively in an epidemic investigation team.
- decide on the success of epidemic management with data from laboratory monitoring.
- critically interpret the findings resulting from the epidemiological surveillance systems and the findings of the investigation of epidemics, to recognize their limitations and to indicate targeted measures for the control and prevention of diseases resulting from them.

General Competencies

Taking into account the general competencies that the graduate must have acquired (as these are listed in the Diploma Supplement and listed below) which of them is the aim of the course?

Search, analyze and synthesize data and information, using the necessary technologies	Project planning and management
Adapting to new situations	Respect for diversity and multiculturalism
Decision-making	Respect for the natural environment
Independent work	Demonstration of social, professional and ethical responsibility and gender sensitivity
Teamwork	Criticism and self-criticism
Working in an international environment	Promotion of free, creative and inductive thinking
Working in an interdisciplinary environment	
Generating new research ideas	

- Search, analyze and complex data and information, using the necessary technologies
- Adapting to new situations
- Decision-making
- Autonomous and teamwork
- Work in an international and interdisciplinary environment
- Generating new research ideas
- Plan and manage projects
- Criticism and self-criticism
- Promotion of free, creative and inductive thinking

3. COURSE CONTENT

Modules:

1. Basic principles of epidemiological surveillance: role in public health
2. Epidemiological surveillance systems in Greece
3. Epidemiological surveillance in emergency situations
4. Design and evaluation of an epidemiological surveillance system
5. Monitoring the evolution of a pandemic: COVID-19, January 2020-December 2021
6. New approaches to epidemiological surveillance. The role of the laboratory in epidemiological surveillance: standardization of pathogens and whole genome sequencing
7. Basic principles of epidemic investigation, organizational issues and required collaborations
8. Epidemics: characteristics, types, descriptive epidemiology (place-time-characteristics of a person)
9. Analytical epidemiology studies in the context of epidemic investigation
10. Methodological issues (random and systematic error) in the context of epidemic investigation

11. Epidemic investigation exercise – steps, organizational issues, collaborations. Critical interpretation of findings
12. Epidemic investigation exercise - cohort study. Critical interpretation of findings.
13. Epidemic investigation exercise - case-control study. Critical interpretation of findings.

4. TEACHING AND LEARNING METHODS - EVALUATION

DELIVERY METHOD <i>Face to face, Distance learning, etc.</i>	Face-to-face teaching (in the classroom), tele-education (synchronous via MS Teams and asynchronous through Moodle), practical exercises on paper, discussion of selected articles										
USE OF INFORMATION AND COMMUNICATION TECHNOLOGIES <i>Use of ICT in Teaching, Laboratory Education, Communication with students</i>	● Use of Power Point in all lectures. ● Posting course material (notes, lecture slides, exercises, exam topics, etc.) on the e-learning platform (Moodle). ● Use of e-mail and announcements on the e-learning platform to communicate with students.										
TEACHING ORGANIZATION <i>The way and methods of teaching are described in detail.</i> <i>Lectures, Seminars, Laboratory Exercise, Field Exercise, Study & analysis of bibliography, Tutorial, Practical (Placement), Clinical Practice, Art Laboratory, Interactive teaching, Educational visits, Study preparation (project), Writing of papers / assignments, Artistic creation, etc.</i> <i>The student's study hours for each learning activity are listed as well as the hours of non-guided study so that the total workload at semester level corresponds to ects standards</i>	<table border="1"> <thead> <tr> <th>Activity</th><th>Semester Workload</th></tr> </thead> <tbody> <tr> <td>Lectures</td><td>26</td></tr> <tr> <td>Laboratory Exercise</td><td>13</td></tr> <tr> <td>Independent Study</td><td>111</td></tr> <tr> <td>Course Total (25 hours of workload per credit unit)</td><td>150</td></tr> </tbody> </table>	Activity	Semester Workload	Lectures	26	Laboratory Exercise	13	Independent Study	111	Course Total (25 hours of workload per credit unit)	150
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STUDENT EVALUATION <i>Description of the evaluation process</i> <i>Assessment Language, Assessment Methods, Formative or Summative, Multiple Choice Test, Short Answer Questions, Essay Development Questions, Problem Solving, Written Assignment, Report/ Report, Oral Examination, Public Presentation, Laboratory Thesis, Clinical Examination of the Patient, Artistic Interpretation, Other / Other</i> <i>Explicitly defined evaluation criteria are mentioned and if and where they are accessible to students.</i>	Written final exam (100%) which includes: – Short answer questions – Problem solving – Multiple-choice questions For successful completion, a grade of at least 5/10 is required										

5. COMPONENT-BIBLIOGRAPHY

- Suggested Bibliography:

1. Rasmussen SA and Goodman RA, editors. The CDC Field Epidemiology Manual. New York: Oxford University Press. 2019. (Selected chapters). Available for free at: <https://www.cdc.gov/eis/field-epi-manual/index.html>.
2. Lee L, Teutsch S, Thacker S, Louis M. Principles and Practice of Public Health Surveillance. 3rd edition. Oxford: OUP. 2010.

3. Michael B. Gregg. Field Epidemiology. 3rd edition. Oxford: OUP. 2008.
4. Bonita R, Beaglehole T, Kjellstrom T. Basic Epidemiology. Editions E.C. Paschalidis. 2009. (Translation 2nd English edition, WHO, 2006: https://apps.who.int/iris/bitstream/handle/10665/43541/9241547073_eng.pdf?sequence=1&isAllowed=y).
5. Rothman KJ. Epidemiology. An introduction. 2nd edition. Oxford: OUP. 2012.
6. Giesecke J. Modern infectious disease epidemiology. 3rd edition. London: CRC Press. 2017.
7. M'ikanatha NM et al, editors. Infectious disease surveillance. 2nd edition. Wiley-Blackwell, 2013.
8. Dworkin MS. Outbreak Investigations Around the World: Case Studies in Infectious Disease Field Epidemiology. 1st edition. Jones & Bartlett. 2010.
9. The Independent Panel for Pandemic Preparedness & Response. COVID-19: Make it the Last Pandemic. May 2021. Available for free at: https://theindependentpanel.org/wp-content/uploads/2021/05/COVID-19-Make-it-the-LastPandemic_final.pdf; <https://theindependentpanel.org/mainreport/>.